UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SOFTWARE DESIGN AND ARCHITECTURE (SDA)

Chapter 4

Innovating Solutions

UTMSPACE ESTD 1993 (931427-K)

01 **INTRODUCTION**

Intro to Architectural Style & Pattern

02 **THE OLD ARCHITECTURAL**

Mapping System, Pattern and Quality

03 **THE NEW ARCHITECTURAL**

Monolithic, Services, and etc..

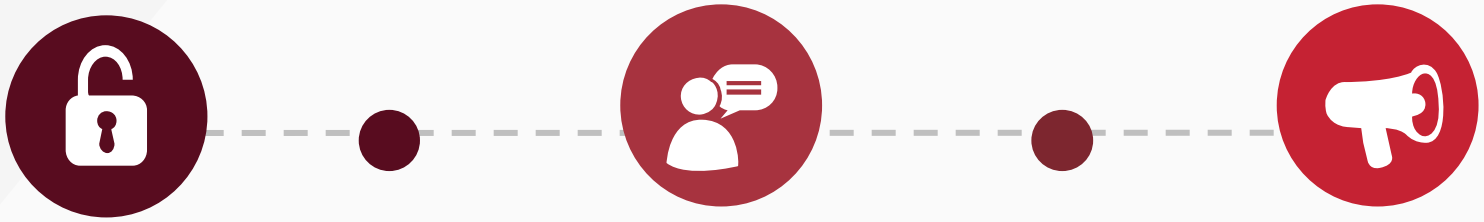
04

05

06

SOFTWARE ARCHITECTURAL STYLE AND PATTERN

Architectural styles and architectural patterns provide generic, reusable solutions that can be easily understood



Generic View

Standard diagrams with details

Solutions

Completeness of solutions

Understood

Easy to understand

THE SYSTEM TYPE

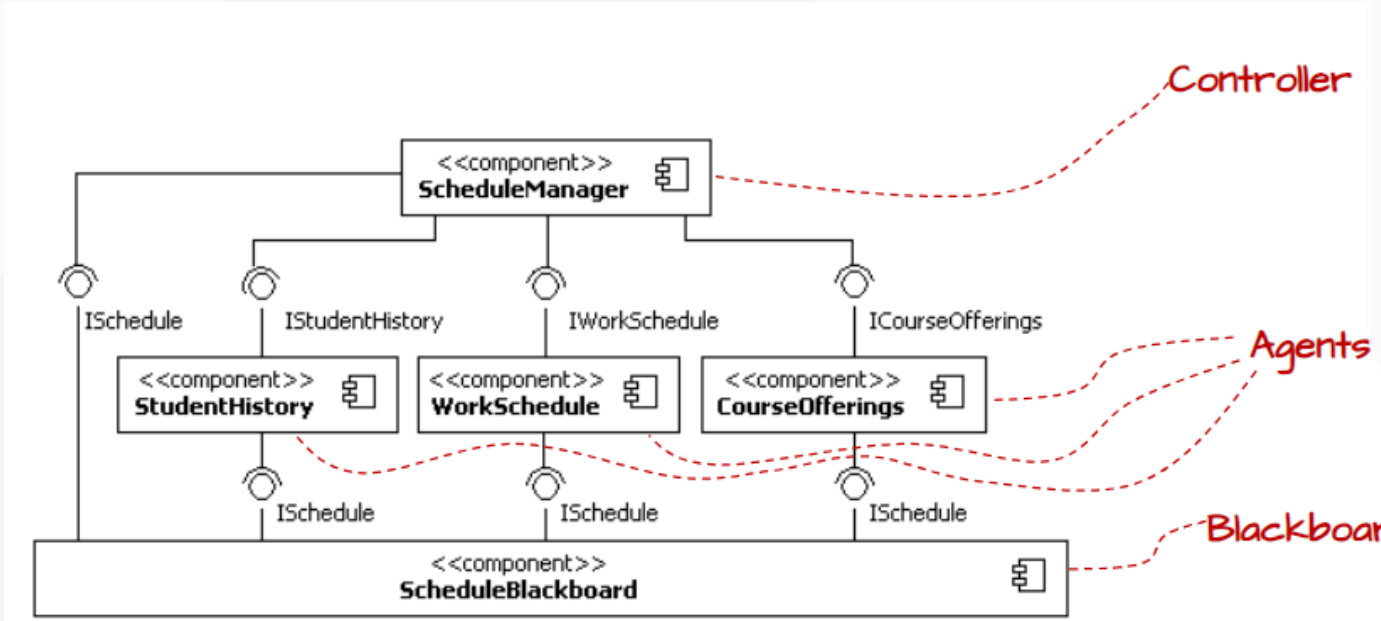
Type	Description
Data-Centered	Systems that serve as a centralized repository for data, while allowing clients to access and perform work on the data.
Data Flow	Systems oriented around the transport and transformation of a stream of data.
Distributed	Systems primarily involve interaction between several independent processing units connected via a network.
Interactive	Systems that serve users or user-centric systems.
Hierarchical	Systems where components can be structured as a hierarchy (vertically and horizontally) to reflect different levels of abstraction and responsibility.

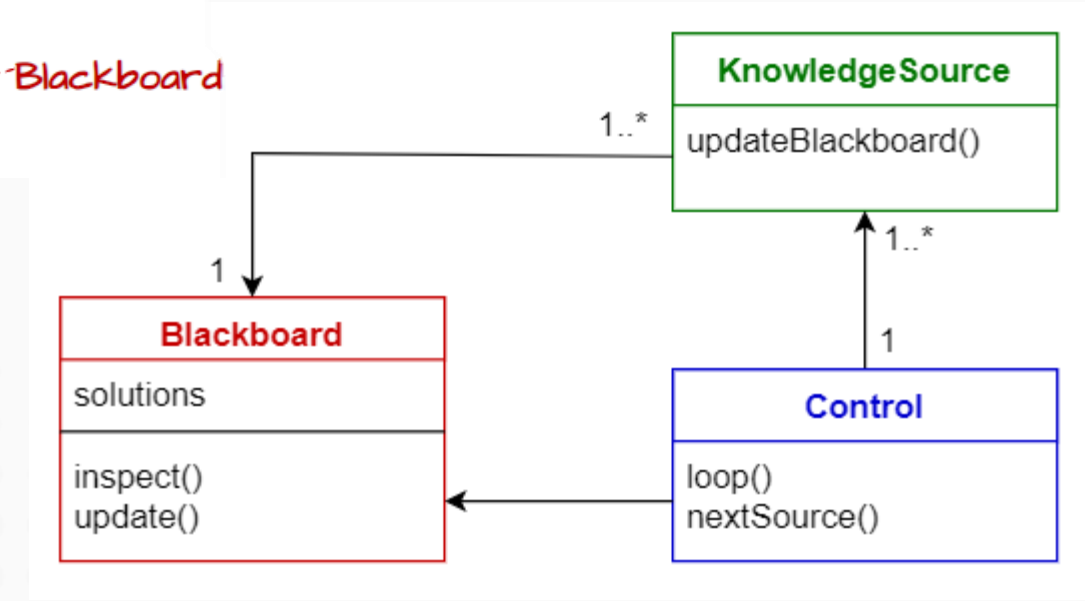
No	System Type	Description
1	Real-Time System	Real-time systems are designed to respond to events or inputs within a specified timeframe, often with stringent timing constraints.
2	Embedded System	Embedded systems are specialized computing systems designed to perform specific tasks within larger systems or devices. They are typically embedded into other devices and often operate in real-time with limited resources.
3	Adaptive System	Adaptive systems have the capability to change their behavior or structure based on internal or external conditions. They can learn from experience, adjust to new inputs, and optimize their performance over time

THE OLD ARCHITECTURAL

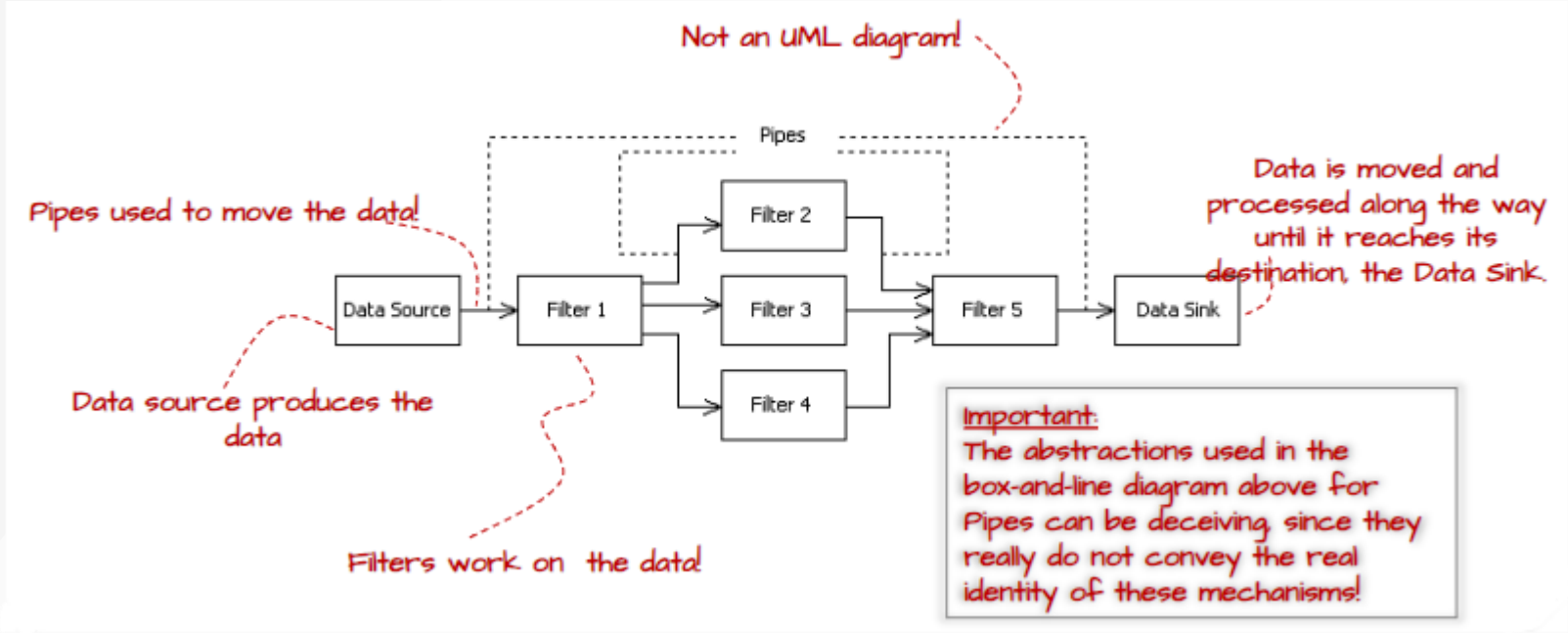
No	System Type	Architecture Pattern	Quality Attribute
1	Data centred system	Blackboard	Modifiability Maintainability Reusability
2	Data flow system	Pipes and Filters	Efficiency Modifiability Extensibility Maintainability Reusability Security
3	Distributed system	Client and Server Broker	Interoperability Modifiability Availability Reusability
4	Interactive system	Model View Controller (MVC)	Modifiability Usability Reusability
5	Hierarchy system	Layered	Modifiability Portability Reusability Security

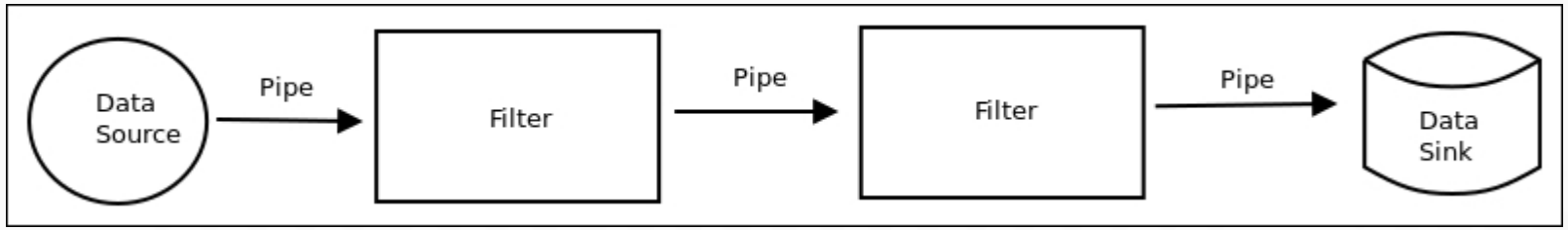
DATA CENTRED – BLACKBOARD ARCHITECTURAL PATTERN



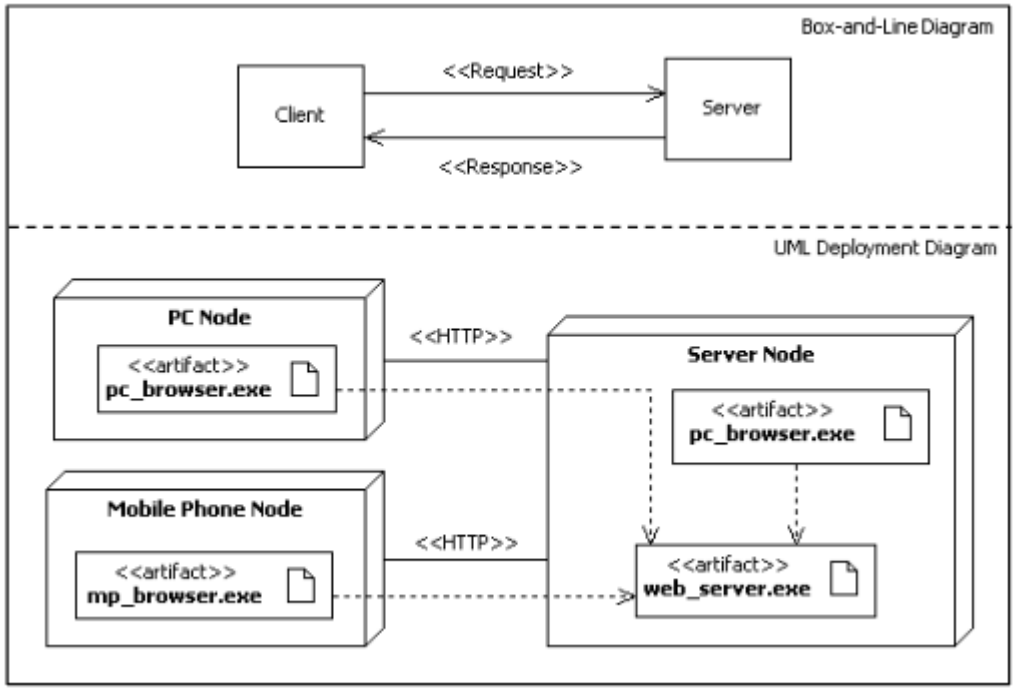


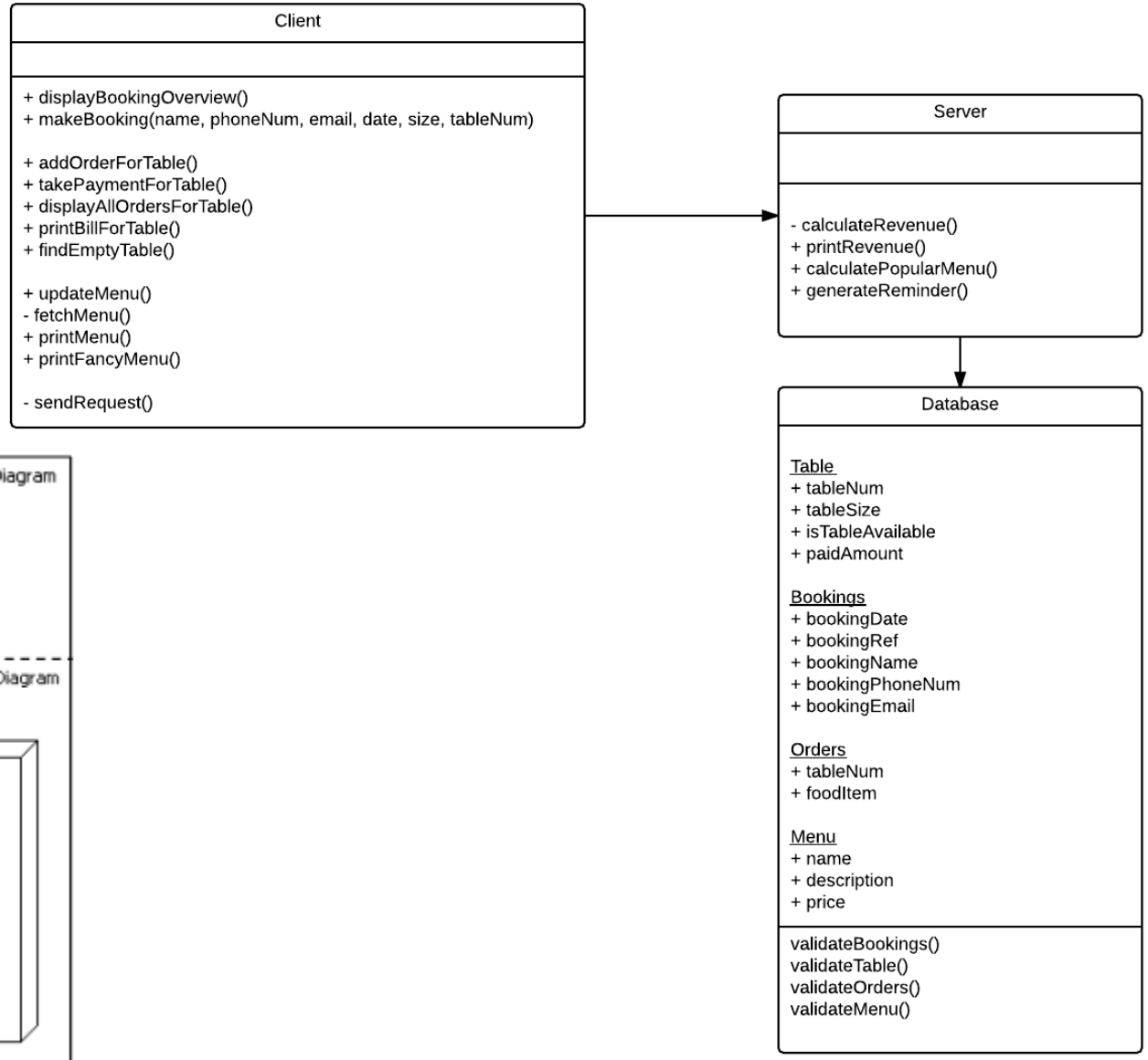
DATA FLOW – PIPES-AND-FILTERS ARCHITECTURAL PATTERN



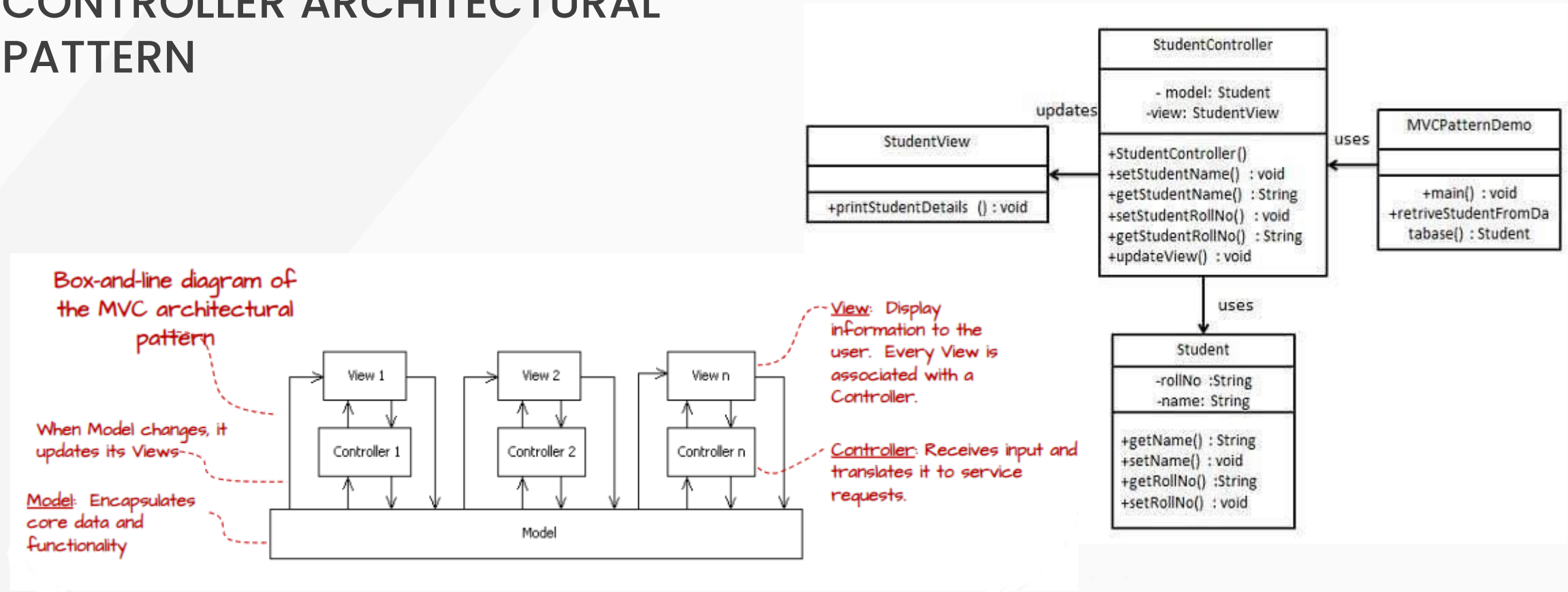


DISTRIBUTED – CLIENT AND SERVER ARCHITECTURAL PATTERN

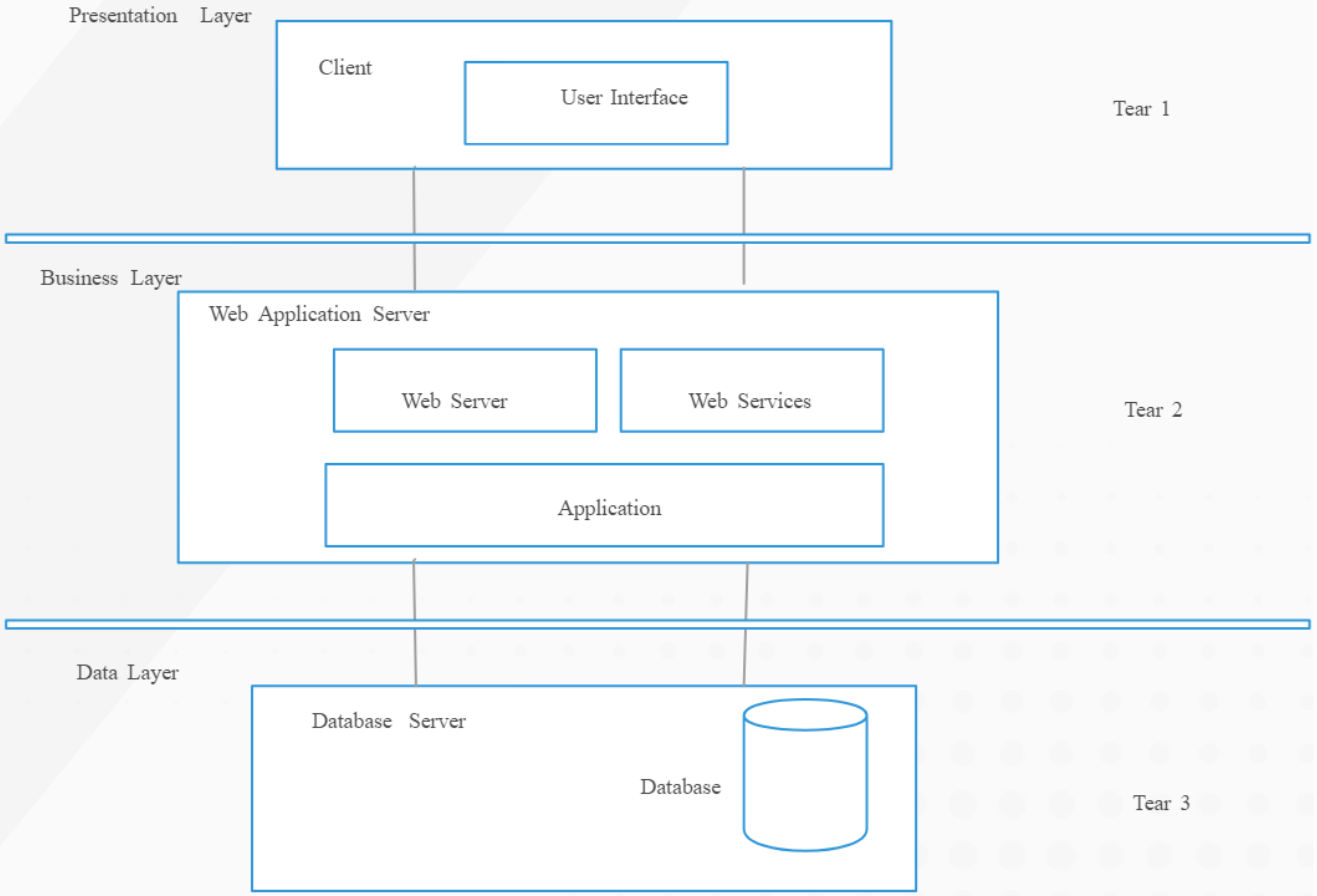


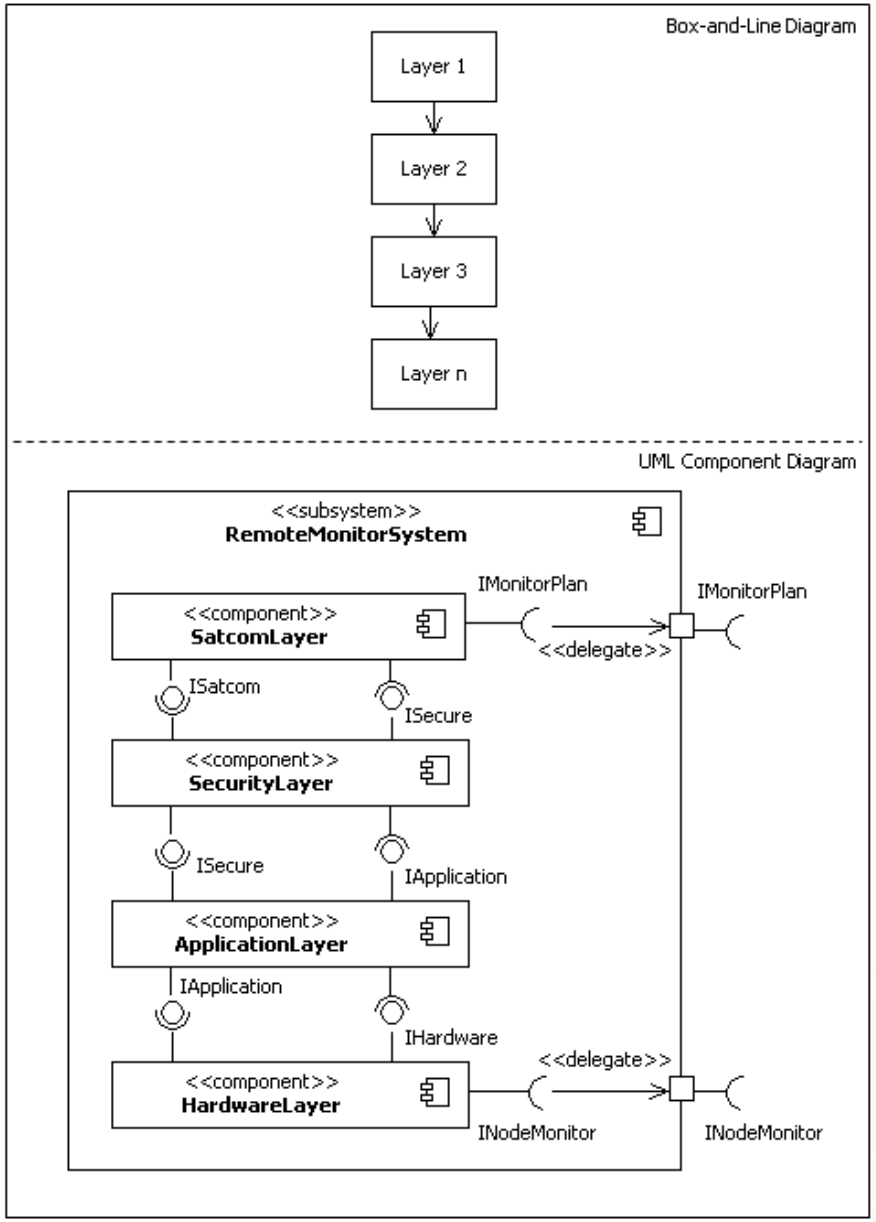


INTERACTIVE – MODEL VIEW CONTROLLER ARCHITECTURAL PATTERN



HIERARCHY – LAYERED ARCHITECTURAL PATTERN





THE NEW ARCHITECTURAL

No	System Type	Architecture Pattern	Quality Attribute
1	Distributed system + Interactive system + Component based	Service Oriented Architecture (SOA)	Interoperability Modifiability Maintainability Availability Reusability
2	Data flow system + Distributed system + Interactive system + Component based + Web Services	Microservices Architecture Serverless Architecture Event-Driven Architecture Hybrid Architecture	Efficiency Modifiability Extensibility Maintainability Reusability Security
5	Hierarchy system + Interactive system + Component based	Modular Monolith Architecture	Modifiability Portability Reusability Maintainability Availability Reusability Security



Thank you for your participation. I am very grateful to have the opportunity into this sharing session. If there any enquiries, feels free to contact me.

+013 509 9215

khatibsyarbini@utm.my

THANK YOU

UTM
UNIVERSITI TEKNOLOGI MALAYSIA

In the Name of God for Mankind

utm.my univteknologimalaysia utmofficial